

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

ANSWERS

CG

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INDEX NUMBER

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BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

12 September 2018

Candidates answer on the Question Paper.

2 hours

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, CG and index number on page 1, 5, 9 and 13 in the spaces provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	15
2	20
3	15
4 or 5	25
Total	75

This document consists of **18** printed pages.



Section A

Answer **all** the questions in this section.

- 1 Red blood cells are formed from cells called reticulocytes. Stem cells in the bone marrow produce reticulocytes which differentiate into red blood cells. Red blood cells have an average life span of 120 days, after which they are destroyed in the liver.

(a) Describe how stem cells in the bone marrow produce red blood cells.

1. asymmetrical division

1 remain SC, the other becomes (myeloid) progenitor cell ;

2. which further divides to give rise to erythroid lineage

which produces reticulocytes that differentiates into RBCs ;

[2]

New red blood cells are constantly produced in the bone marrow as stem cells, unlike other somatic cells, can divide throughout the lifespan of the individual. Somatic cells undergo a limited number of cell divisions as the DNA daughter strand is shortened with every replication.

Fig. 1.1 shows the shortened daughter strand with its parent template after a replication.



Fig. 1.1

(b) (i) Explain why the daughter DNA strand is shortened after each replication.

1. end replication problem

DNA pol can only elongate a pre-existing strand w a free 3'OH end ;

2. no preceding strand exist after removal of the last / terminal primer at the 5' end of the daughter strand

thus a gap occurs in the daughter strand, with a 3' overhang in the parent strand ;

[2]

(ii) Explain why stem cells are able to divide throughout the lifespan of the individual.

1. due to presence of active telomerase

that lengthens parent strand (by several tandem repeats) ;

2. to (act as template for) primase to synthesize another primer further upstream for DNA pol (III) to synthesize daughter strand that is as long as the initial parent strand ;

[2]

During differentiation, haemoglobin is produced. Fig. 1.2 shows the structure of a section of the haemoglobin gene and its corresponding messenger RNA (mRNA) in the nucleus of a reticulocyte during transcription.

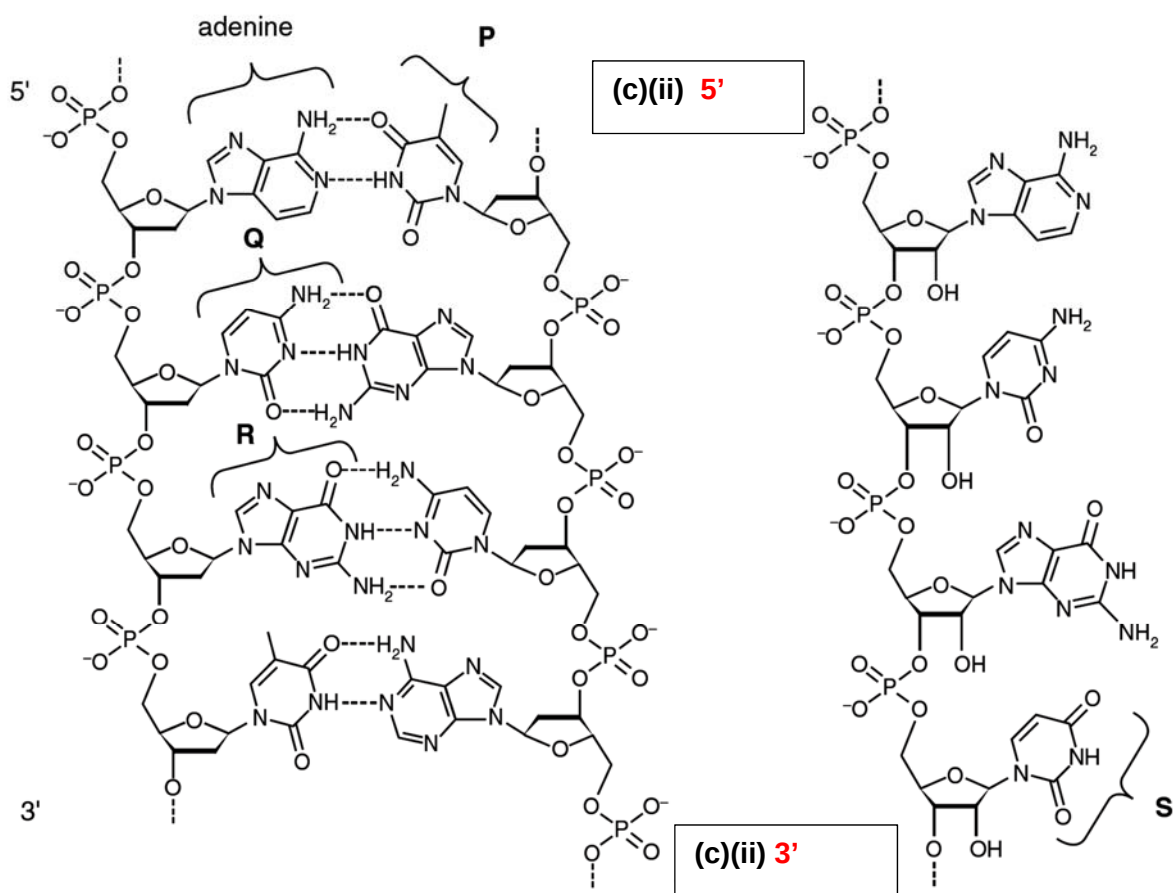


Fig. 1.2

(c) (i) Identify structures P – S.

P *thymine*

Q *cytosine*

R *guanine*

S *uracil*

[2]

(ii) On Fig. 1.2, label the 5' and 3' end of the mRNA molecule.

[1]

- (iii) Briefly describe the fate of the mRNA molecule after transcription before its export out of the nucleus.

1. 5' capping

addition of methylated guanine (residue) at 5' end ;

2. 3' polyadenylation

addition of multiple adenine residues/nucleotides at 3' end (after polyadenylation signal) ;

3. splicing

removal of introns, joining of exons (by spliceosome) ;

[3]

- (d) Describe **two** differences between transcription and DNA replication.

1. template

2 DNA parent strand vs 1 DNA template / nonsense strand ;

2. product

(two) DNA molecules (each consisting of 1 parent coiled w 1 daughter strand) vs RNA / uses deoxyribonucleotides vs ribonucleotides ;

® thymine vs uracil ® rRNA vs 1 daughter strand / 2 daughter strand (implies 2 daughter strands coiled together)

3. enz

DNA pol vs RNA pol ;

[2]

® primase / helicase / topoisomerase vs none

- (e) Suggest **one** additional change, besides production of haemoglobin that occurs to produce reticulocytes during their differentiation into red blood cells.

1. degradation of nucleus

(by hydrolytic enz in lysosome / fusion w lysosome / autophagy) ;

[1]

[Total: 15]

- 2 Lactate dehydrogenase (LDH) is an enzyme found in many organisms. Within the same organism, it can be found in different forms, called isoenzymes. The isoenzymes are structurally different but they all catalyse the same reaction.

Fig. 2.1 shows a reaction catalysed by lactate dehydrogenase that occurs during anaerobic respiration in muscle tissue.

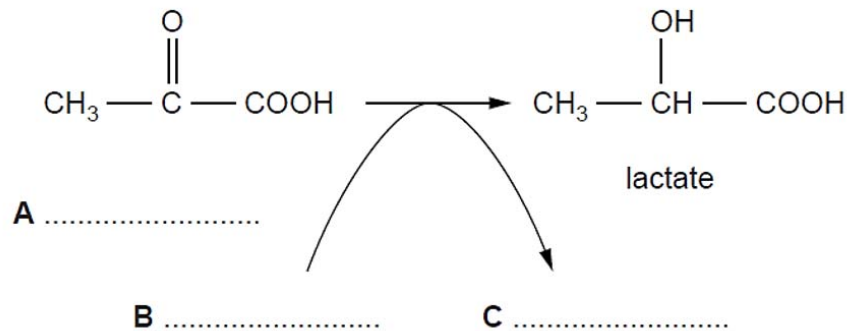


Fig. 2.1

(a) With reference to Fig. 2.1,

- (i) identify compounds A, B and C.

A **pyruvate;**

B **reduced NAD/ NADH + H⁺;**

C **oxidised NAD/ NAD⁺;**

[2]

- (ii) state the location in the cell where this reaction takes place.

cytosol;

[1]

- (iii) explain the significance of this reaction in mammalian muscle tissue.

1. pyruvate reduced to lactate

In absence of O₂ in anaerobic respiration;

2. where NADH is oxidised to NAD

allows NAD regeneration for glycolysis to continue;

3. allows ATP pdtn for glycolysis to continue for muscle contraction

Lactate converted back into pyruvate & glucose in liver;

[3]

Lactate dehydrogenase isoenzymes are globular proteins, each consisting of four polypeptides.

(b) Explain how the structure of an enzyme, such as lactate dehydrogenase is suited to its role.

1. has 4^o prot struc (4 polypeptides)

each polypeptide has 3^o struc held in place by bds/ interxns b/w R grps of aas;

2. bonds/ interxns (e.g. H bds, hydrophobic interxns) b/w polypeptides

forms specific 3D config with active site compl to substrate;

3. active site is compl (shape, size, charged, orientation)

brings catalytic residues nearer each other in active site;

4. ES complex formation lowers E_a for catalysis

bond angle (stress bonds)/ favourable microenv/ stabilize transition state;

5. globular prot with hydrophilic aas on exterior allowing prot to be soluble

rxns in aqueous medium/ ref to tpt of enz;

[4]

Lactate dehydrogenase isoenzymes are made up of two types of polypeptide: polypeptide M, which is coded for by the *LDH-A* gene and polypeptide H, which is coded for by the *LDH-B* gene.

Table 2.1 shows the composition of different human lactate dehydrogenase isoenzymes and examples of tissues and organs where each can be found.

Table 2.1

isoenzyme	polypeptide composition of enzyme	example of isoenzyme location
LDH-1	HHHH	heart red blood cells
LDH-2	HHHM	heart red blood cells
LDH-3	HHMM	brain lungs
LDH-4	HMMM	kidneys placenta
LDH-5	MMMM	liver skeletal muscles

(c) With reference to Table 2.1, suggest how red blood cells of the same individual can produce different isoenzymes.

1. genome of rbc progenitor cells contains both genes that code for LDH-A & LDH-B polypeptides;

2. diff expression of genes at diff times & locations of cell pdc diff isoenzymes (QV frm Table 2.1); OR

3. ctrl of assembly of polypeptides to pdc diff prots/ diff polypeptide chains make up diff prots;

4. diff R grp interxns b/w diff polypeptide chains, resulting diff 4^o struc/ specific 3D config (QV frm Table 2.1);

[2]

Besides lactate dehydrogenase, another extensively studied protein which is also involved in respiration is cytochrome c. Cytochrome c plays an important role in oxidative phosphorylation as an electron carrier of the electron transport chain.

Table 2.2 shows the amino acid sequence of a section of the cytochrome c polypeptide chain retrieved from 1 individual from each of the species below. The dashes shown in the figure indicates that the amino acid present at the position is identical to that of the human species.

Table 2.2

		1					6				10				14			17	18		20		
Human		Gly	Asp	Val	Glu	Lys	Gly	Lys	Lys	Ile	Phe	Ile	Met	Lys	Cys	Ser	Gln	Cys	His	Thr	Val	Glu	Lys
Pig		-	-	-	-	-	-	-	-	-	-	Val	Gln	-	-	Ala	-	-	-	-	-	-	-
Chicken		-	-	Ile	-	-	-	-	-	-	-	Val	Gln	-	-	-	-	-	-	-	-	-	-
Dogfish		-	-	-	-	-	-	-	-	Val	-	Val	Gln	-	-	Ala	-	-	-	-	-	-	Asn
Drosophila	<<<	-	-	-	-	-	-	-	-	Leu	-	Val	Gln	Arg	-	Ala	-	-	-	-	-	-	Ala
Wheat	<<<	-	Asn	Pro	Asp	Ala	-	Ala	-	-	-	Lys	Thr	-	-	Ala	-	-	-	-	-	Asp	Ala
Yeast	<<<	-	Ser	Ala	Lys	-	-	Ala	Thr	Leu	-	Lys	Thr	Arg	-	Glu	Leu	-	-	-	-	-	-

(d) With reference to Table 2.2,

(i) suggest what information indicates about the evolutionary relationships between humans and the other species;

1. all spp share a common ancestor/ prot is highly conserved;

2. smaller the no. of diff^e more closely related/more recently divergence occurred/ lesser time to accumulate diff mutations;

3. pigs & chickens are phylogenetically closest to humans with only 3 no. of aas diff from humans

yeast is most distantly related to humans with 11 aas diff from humans;

[3]

(ii) explain why the stated conclusion in (d)(i) needs to be treated with caution.

1. c.f. aa seq (22 aas) is only coded by a small fraction of whole genome;

2. cyt c is likely to be a larger polypeptide/ prot, hence not good representation; OR

3. aa seq in Table 2.2 comes from 1 indiv of each spp, should be variation within each spp;

4. small sample size/ use of aa seq does not allow detection of silent mutations due to degeneracy of genetic code; OR

5. aa seq only reflects coding region of DNA seq;

6. changes in non-coding regions are not detected as these spp are eukaryotic, hence larger non-coding regions;

[2]

(e) Suggest how the differences in the amino acid sequences shown in Table 2.2 may have arisen.

1. point/ missense mutation in nucleotide seq causing change in single base via substitution; OR

mis-pairing of compl bases (e.g. pyrimidine dimers) during DNA replication;

2. by mutagens e.g. UV, X-rays, ionising radiation;

[2]

- (f) Suggest why the cytochrome c protein was chosen to compare amino acid sequences across the different species.

1. *cyt c prot is ubiquitous/ present in all org as all org need to respire hence serves as a good comparison b/w org; OR*

cyt c gene is an essential gene that changes very slowly, useful for estimating time of divergence that occurred long time ago; [1]

[Total: 20]

- 3 A student cut thin sections of a root tip of *Allium cepa* (*A. cepa*) and stained them to show chromosomes. A photomicrograph of part of one of these sections is shown in Fig. 3.1.

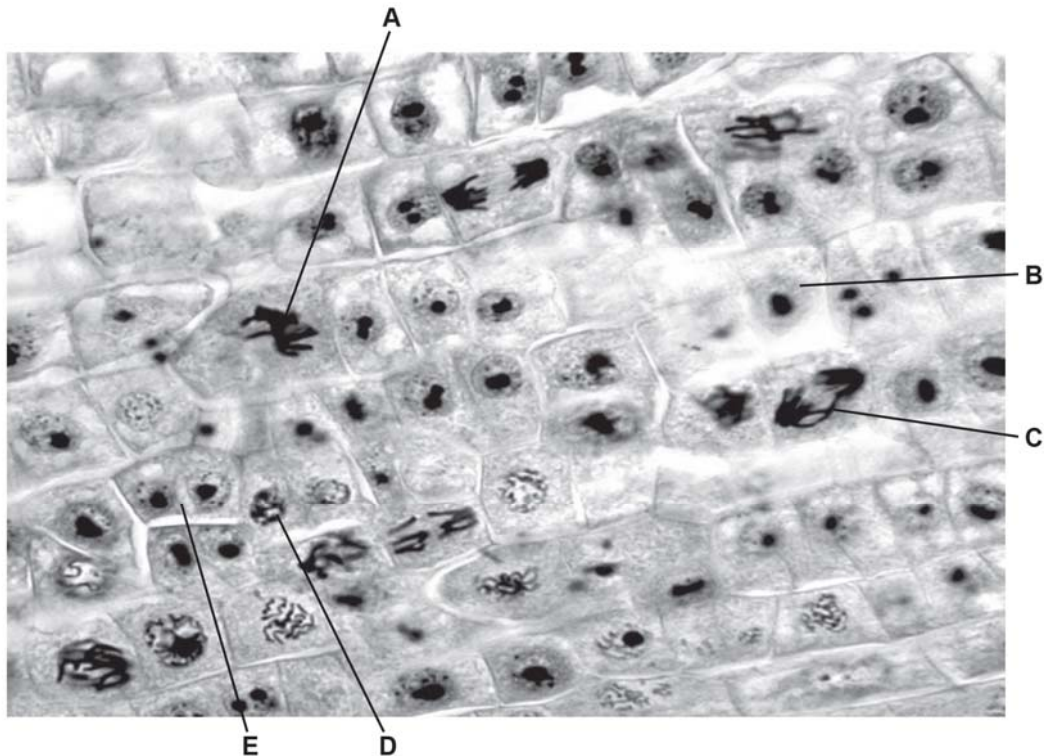


Fig. 3.1

- (a) Table 3.1 shows the behaviour of chromosomes and the changes that occur to the nuclear envelope during a mitotic cell cycle in the root tip of *A. cepa*.

Complete Table 3.1.

Table 3.1

name of stage	letter of cell	behaviour of chromosomes	nuclear envelope
<i>interphase</i>	B	uncondensed, may be replicating	<i>present / intact</i>
prophase	<i>D</i>	<i>condensed, visible as chromosome / 2 sis chromatids held at centromere</i>	intact, eventually breaks down
metaphase	<i>A</i>	<i>aligned (in a single row) at equatorial / metaphase plate ® equator alone, middle of cell</i>	Not present
anaphase	<i>C</i>	<i>move to oppo / spindle poles of cell ® oppo ends of cell</i>	Not present
telophase	<i>E</i>	uncoil / to become thin chromatin threads	<i>reforming / present / intact</i>

- (b) Explain why the growth of roots, such as those of *A. cepa*, involves mitosis and not meiosis.

1. mitosis produces genetically identical

diploid daughter cells / maintain genetic stability ;

2. meiosis produces genetically varied / non-identical

haploid daughter cells ; (l) produce gametes

3. growth of roots requires similar cells (to be grouped together)

to form tissue / organ / perform same fn / replace damaged cells ;

® repair damaged cells

[3]

- (c) Some bacteria infect the plant to result in tumour formation at the infected tissues as shown in Fig. 3.2.

Research on the tumour formation processes revealed that there was an unusual excessive production of auxin, an important plant growth hormone. Auxin has been found to regulate gene expression resulting in cell division, expansion, differentiation and specialisation.

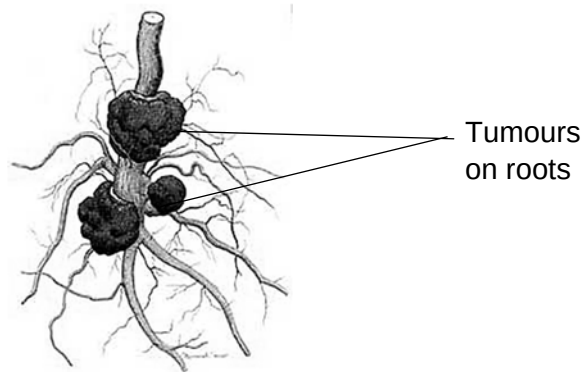


Fig. 3.2

Suggest how the bacteria infection resulted in tumour formation.

1. bacteria triggers infected plant cell to produce excessive auxin

➡ uncontrolled cell proliferation / dysregulation of cell cycle / G1 chkpt

2. auxin upregulates / ↑ expression / transcription

of POG / genes coding for production of CDKs / cyclins / prot that ⊕ cell division ;

3. auxin activates telomerase gene ➡ active telomerase

allowing plant cells to divide indefinitely / unlimited replicative potential ;

4. auxin downregulates / ↓ expression / transcription

of TSG / genes coding for prot involved in apoptosis / cell cycle arrest ;

[3]

- (d) A protein, mitosis promoting factor (MPF), has been identified in cells and has a role in initiating prophase.

MPF is made up of cyclin B1 coupled with cyclin-dependent kinase 1 (CDK1) and normally begins to break down and stops functioning during anaphase.

An overexpression of cyclin B1 has been reported in various human tumours, and is related to aneuploidy and high growth rate for cancerous growths in the breast.

- (i) Suggest reasons for the overexpression of cyclin B1.

1. state at least 1 cause of mutation

e.g. DNA replication errors / named mutagen / category of mutagens

2. cause chromosome aberration ➔ cyclin B1 gene is amplified / duplicated

expression of multiple copies of cyclin B1 gene ➔ ↑ cyclin B1 production ;

3. cause gene mutation / Δ in DNA seq of cyclin B1 gene

➔ Δ in 3D config. of cyclin B1 ➔ not degraded by degradative enz ;

4. cause gene mutation / Δ in DNA seq in gene regulating cyclin B1 gene expression / core promoter / enhancer

➔ ↑ transcription of cyclin B1 gene ;

[3]

- (ii) State what does aneuploidy mean.

abnormal / extra / reduction / Δ in no. of chr ;

[1]

® abnormal no. of sets of chromosome e.g. 3n / 4n etc.

® ref. to human ploidy e.g. has 45 / 47 chromosomes

[Total: 15]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

- 4 (a) ATP is known as the universal currency of energy in living organisms. It can be utilised and regenerated in various chemical and physical reactions that occur in living systems.

With reference to named examples, outline the roles of ATP in living organisms. [13]

General

1. *small, easily transported, water soluble
energy currency for cellular processes ;*
2. *high energy bonds between phosphate groups
when broken releases energy / exothermic reaction ;*

Transport & Arrangement

3. *reestablishment of ion concentration gradient in neurones
Na⁺/K⁺ pump to move ions against concentration gradient;*
4. *movement of vesicles along cytoskeleton
provide e.g. between organelles / to CSM (secretion of insulin from pancreatic cells/antibodies from plasma cells);*
5. *deformation of membrane for bulk transport
endocytosis/exocytosis/receptor mediated endocytosis/phagocytosis ;*
6. *assembly of microtubules in spindle fibres
arrangement of chromosomes during cell division;*

Cell Signalling

7. *substrate for adenyl cyclase
convert to cAMP → 2nd messenger for cell signalling e.g. G-protein signalling cascade/ lac operon in bacteria*
8. *activation of RTK through autophosphorylation
docking ports for relay proteins ;*

Energetics**9. involved in glycolysis**

transfer of phosphate from ATP to activate glucose (or give e.g.);

10. concentration of ATP regulates glycolysis

allosteric inhibitor to PFK ;

11. used in PGA reduction reactions in Calvin Cycle

transfer of phosphate from ATP to 3-PGA → 1,3- Bisphosphoglycerate

12. used in regeneration reactions in Calvin Cycle

G3P → RUBP

Other cellular processes**13. tRNA activation by amino-acyl tRNA synthase**

aa binding to active site of enzyme

14. synthesis of macromolecules

aa synthesis/polysacc/lipid synthesis ;

15. substrate for RNA synthesis by RNA polymerase

tRNA, rRNA, mRNA ;

Additional e.g.**16. Additional e.g. for categories mentioned above (e.g. phosphorylation of relay proteins with e.g.)****17. Additional e.g. for categories mentioned above (e.g. active transport of materials against concentration gradient with e.g.)****18. Additional e.g. for categories mentioned above****19. AVP**

- **QWC (1m): points communicated clearly without ambiguity & with relevant e.g.**

- (b) Plasma membranes exist on the surface of cells and in the endomembrane systems within the cells. They play critical roles in the regular functioning of a cell.

Discuss the functions of various components in plasma membrane and explain why there is a different composition of these components in membranes of different cells and organelles. [12]

Components of membrane and their functions:

Phospholipids

1. **Phospholipids, amphipathic**
forms bilayer ;
2. **acts as barrier**
hydrophilic/ water-soluble substances ;
3. **held by weak hydrophobic interactions**
provides fluidity to memb ;

Cholesterol

4. **cholesterol regulates fluidity**
and permeability of membrane
5. **keeps phospholipids together at high temperature**
to prevent membrane breakage ;
6. **prevents phospholipids from packing at low temperature**
prevents membrane solidification ;
7. **function as a plug**
regulate movement of molecules aX membrane ;

Proteins

8. **transport proteins**
allow hydrophilic molecules to be tpted in & out of cell / aX membrane
9. **enzymes**
catalyse chemical rxns on memb
10. **receptor**
allows for specific binding of ligand/ 1st messenger in cell signalling pathways
11. **structural support**
prots attached to cytoskeleton to provide framework for cell ;
12. **energy transducers**
harness energy from proton flow to phosphorylate ADP e.g. ATP synthase ;

Carbohydrates (@glycoproteins or glycolipids)

13. **form H bonds with water**
& stabilizes memb
14. **on proteins and lipids forming glycoproteins and glycolipids**
cell-cell recognition / cell communication / cell-cell adhesion

Significance of different components:

In different organelles (relate to its functions)

Transport proteins

15. *function as electron transporters on ETC to facilitate electron transfer along e.g. OP in mitochondrion & light-dependent rxns in chloroplast ;*
16. *proteins function as hydrogen ion transporter / pump harness energy lost from e^- transfer to move H^+ against conc gradient*
17. *proteins function as nuclear pores regulate the movement of molecules in and out of nucleus*
18. *Proteins function as channel proteins On rER for newly translated proteins to enter lumen ;*
19. *Proteins act as proton pumps Maintain acidic internal env in lysosomes ;*

Others

20. *Proteins complex with pigments in light harvesting complex Maximise efficiency of light dependent reactions ;*
21. *Proteins function as enzyme ATP synthase to transport H^+ from IMS to matrix / thylakoid lumen to stroma*

Phospholipids

22. *phospholipids in internal membranes of organelles (e.g. IMM / thylakoid membrane) allows the setting of a proton gradient for chemiosmosis to take place as H^+ flows down proton grad to produce ATP ;*

In different cells (relate to its functions)

Cell Signalling

23. *diff cell types contains diff amt & types of glycoproteins & glycolipids similar cell types can adhere together to form tissues*
24. *cells receiving signalling ligand / specific e.g. of effector cells etc req specific prot receptors to be present on cell memb*
25. *immune cells e.g. B cell/macrophage/ T helper cell etc contain specific receptor e.g. BCR with specific antigen binding sites to bind to antigen ;*

Transport

26. *Channel/carrier proteins for movement of molecules from ECF into cell for storage/processing Give e.g. GLUT on liver/muscle cells*

Others

- 27. higher amount of cholesterol / higher % of saturated fatty acids in phospholipids in membranes of cells in organisms living in areas of higher temp for greater stability of the memb**
- 28. AVP (cell Signalling e.g. recognition of self vs non-self)**
- 29. AVP (Transport)**
- 30. AVP (compartmentalisation of processes)**
- 31. AVP (fluidity of membrane → fluidity and vesicle formation)**

- **QWC (1m): points communicated clearly without ambiguity & with relevant**

[Total: 25]

- 5** The immune system responds to foreign antigens by producing specific immune responses to remove the antigen.

- (a)** Outline the processes involved in the generation of antibody diversity which allowed the humoral immunity to be effective in both the primary and secondary response. [12]

- 1. Antibodies are made up of 4 polypeptides**
Arranged in a Y structure ;
- 2. Each polypeptide is coded for by more than 1 gene segment**
Coding for the variable and constant regions of each polypeptide ;
- 3. Variable regions of the light chains and the heavy chains**
Forms the antigen binding site which is complementary to the antigen epitope
- 4. (in naïve B cells) DNA segments coding for the variable regions**
undergoes genetic recombination ;
- 5. Different coding sequences after genetic recombination**
→ different polypeptides → different antigen binding site ;
- 6. There are 2 types of light chains (κ and λ)**
1 type of heavy chain ;
- 7. Any combination of L and H chain can occur**
With both light chains identical ;
- 8. Inheritance of large number of gene segments for both L and H chains**
Coding for VJ (L chain) and VDJ (H chain) segments ;
- 9. Any combination of V(D)J gene segments can occur**
combinatorial diversification resulting in specific aa sequence for each combination for the antigen binding site ;
- 10. Imprecise joining between gene segments of V to J (L chain) and V to (D)J (H chain)**
Results in junctional diversity in the hypervariable regions in the antigen binding site ;

11. *Upon antigen presentation (in germinal centres), hypermutation of the genes coding for the variable segment and class switching occurs ;*
 12. *Changes nucleotide sequence coding for the hypervariable regions
Altering antigen binding sites of antibodies ;*
 13. *B cells with high affinity antibodies selected
Continue clonal expansion to produce antibodies specific for antigenic epitopes for effective immune response during secondary exposure ;*
 14. *Class switching occurs
Gene rearrangements of the constant regions while retaining the genes of the variable regions ;*
- **QWC (1m): points communicated clearly without ambiguity &with relevant**

- (b) The human immunodeficiency virus (HIV) is a retrovirus that causes AIDS. The infected person has a weakened immune system and is susceptible to opportunistic infections which can lead to their death.

Describe the HIV reproductive cycle and explain how these opportunistic infections may arise. [13]

1. **Adsorption involving gp120**
Binding to CD4 receptor and co-receptor on cell surface of host cell ;
2. **Penetration due to membrane fusion**
exposed fusion domain due to conformation changes in gp41 ;
3. **Nucleocapsid released into cytoplasm**
Capsid degradation results in release of viral enzymes and RNA ;
4. **Reverse Transcription by reverse transcriptase**
dsDNA formed from ssRNA ;
5. **dsDNA enters nucleus through nuclear pore**
integrated into host DNA by integrase ;
6. **Provirus state**
Persist for a long time in the latent state with viral genome replicated along with host's chromosomes ;
7. **Upon activation, viral gene expression occurs**
Using cell's protein synthesis mechanisms ;
8. **mRNA synthesized functions as a template**
for translation and genome replication ;
9. **viral ENV polyproteins synthesized cleaved by cellular protease**
forms gp120 and gp41 which are inserted into the CSM ;
10. **viral polyproteins (GAG-POL) polyproteins are cleaved by viral protease**
forms viral enzymes (integrase, RT, protease) and structural proteins respectively ;
11. **mature virons leave the cell via budding**

CSM taken from host cell becomes the envelope ;

12. *lost of CSM results in destruction of CD4 cells*

population of CD4 T/ T helper cells dy/dx into TH ↓ ;

13. *B cells are not activated to dy/dx into plasma cells by TH*

No antibodies secreted (no humoral immunity) against pathogens ;

14. *CD8 cells not activated to dy/dx into Tc*

Pathogen infected cells are not destroyed (lack cytotoxic mechanisms) ;

15. *Compromised immunity results in opportunistic infections*

By pathogens which usually do not cause serious illnesses in people with functional immune system ;

- ***QWC (1m): points communicated clearly without ambiguity &with relevant***

[Total: 25]